



Automating RPI

From writing code to directing agents





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The hottest new programming language is English.

Andrej Karpathy
(OpenAI Co-Founder, Eureka Labs Founder)

Challenges of the AI Engineering Age



Vibe coding lacks quality and engineering rigor

AI tooling can generate things quickly that appear to work but without the human domain and discipline expertise, it will likely be unsustainable, unscalable, and insecure.

Solving the wrong problem fast

Building solutions faster with AI tooling doesn't matter if we aren't solving problems that drive real business impact.

AI without context and data is guesswork.

Without trusted sources that include reliable code, context, and opinionated prompts, agents won't do their best work. In addition to working from a solid data estate and labeled training data, agents need context to ensure quality, security, and best practices.

Agentic in GitHub





Custom Agents in the IDE

A custom agent is a [.github/agents/](https://github.com/agents/) prompt file that defines a persona, a set of instructions, and optionally a list of tools. Once saved to your repo, it appears automatically in the Copilot Chat agent picker alongside built-in agents.

However, sometimes we might not know where to start. Some ideas can be:

Ticket-to-Pr Agent

Takes a Jira or ADO work item, reads the codebase via MCP, and produces a fully scoped implementation plan before writing a line of code.

Domain Expert Agent

Loaded with knowledge of your specific domain — fintech rules, compliance requirements, internal APIs. Every response is grounded in your context.



What is HVE Core?

Microsoft's open-source prompt library for GitHub Copilot

HVE Core is a collection of purpose-built agents, instructions, and prompts that enforce a research-first discipline in Copilot. Instead of jumping straight to code, agents that:

read → plan → get approval → implement.

This is an Extension for VSCode that imports:

Component	Count	Description	Documentation
Agents	49	Specialized AI assistants for research, planning, and implementation	Agents
Instructions	102	Repository-specific coding guidelines applied automatically	Instructions
Prompts	63	Reusable templates for common tasks like commits and PRs	Prompts
Skills	11	Self-contained packages with cross-platform scripts and guidance	Skills
Scripts	N/A	Validation tools for linting, security, and quality	Scripts

MCP + Custom Agents for RPI



Today we will use the 2 MCPs: the **Github MCP** and the **DevOps MCP**

These will allow us to work on the IDE with PRs, Issues and Discovery tools. To do that, we will work with HVE core's Agents for RPI and connect them to these tools



How it all fits together



Agent Mode

The engine

Gives Copilot the ability to take multi-step autonomous actions — read, edit, run, iterate — across your entire workspace.

MCP Servers

The eyes

Connects the agent to live external data — your actual repos, issues, work items, and PRs — instead of hallucinated structure.

HVE Core

The brain

Enforces a research-first discipline. Agents read before they write, plan before they implement, and pause for your approval.



What is Copilot Cloud Agent in GitHub



- Takes issues → produces pull requests (not just suggestions)
- Works like a virtual developer: fixes, features, tests, docs
- Runs in an isolated GitHub Actions environment with auditability + policies enforced
- Human-approved by design: can't merge; PR goes through normal review



Delegate tasks. Stay in control.

Meet Mission Control

Agents

Delegate tasks to GitHub Copilot coding agent to work on in the background, and then monitor Copilot's progress.

Describe a coding task to work on

vfm2-testing-co/gallery-repo

Auto

Models

- ✓ Auto 1x
- AI Claude Opus 4.6 3x
- AI Claude Sonnet 4.6 1x
- GPT-5.2-Codex 1x
- GPT-5.3-Codex 1x
- GPT-5.1-Codex-Max 1x
- AI Claude Sonnet 4.5 1x
- AI Claude Opus 4.5 3x

Get started with agents

- Explain repository**
Let the agent help you understand the codebase
- Improve test coverage**
Analyze the codebase to improve coverage

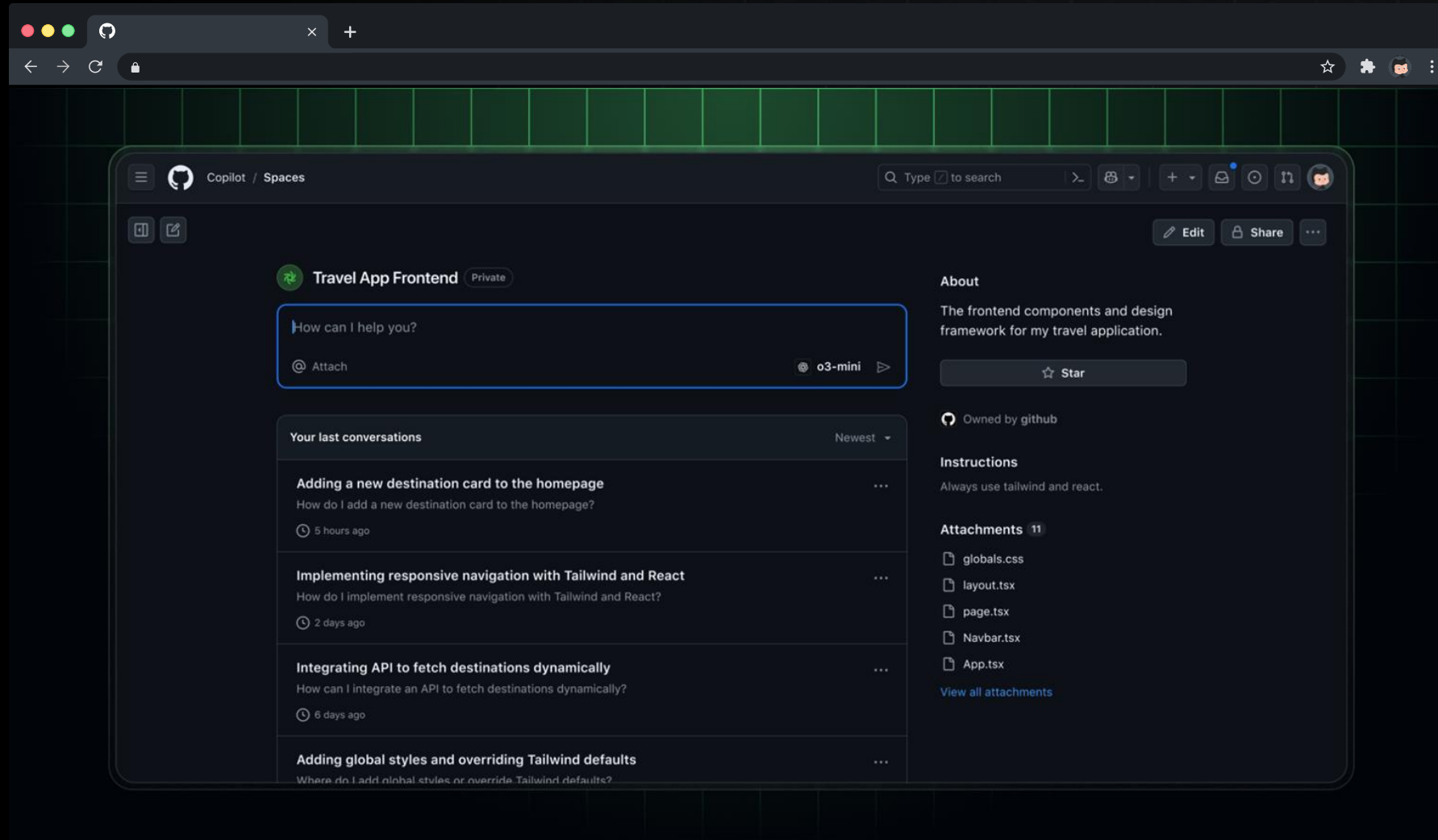
Recent sessions

- Exploring the potential of GitHub Copilot in event**
vfm2-testing-co/gallery-repo #3 · Started 2 weeks ago · Close



Set context once. Chat and chat again

Create and share a custom Copilot experience using a collection of GitHub repos, files, and other content.





Lab 05

RPI Workflow: GitHub MCP, HVE & Coding Agent



Today's Lab

Setup

Connect to the Github or DevOps MCPs

Part 1 — Octocat Demo

Part 1 — Explore the reference repo with MCP and pick a real issue to fix.

Part 2 — Connect Agents to the MCP Server

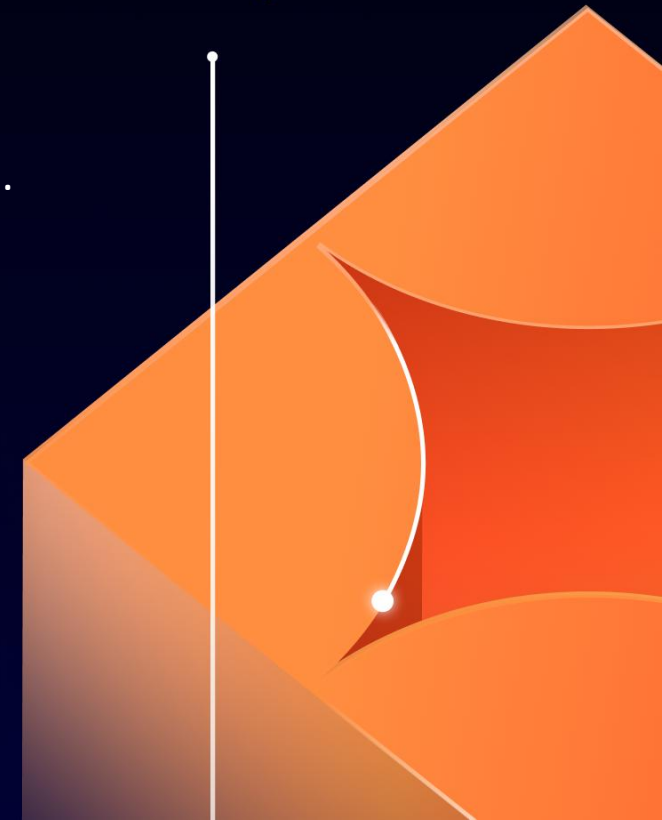
Part 2 — Connect the DevOps/GitHub MCP server and verify agents can see your live repo.

Part 3 — HVE Collection

Run rpi-agent through a full Research → Plan → Implement cycle.

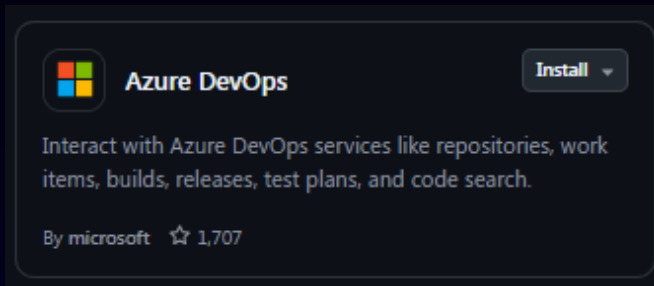
Part 4 — Coding Agent End-to-End

Assign an issue to @github-copilot and review the PR it opens.



Next Steps

Extend the RPI loop into your full toolchain



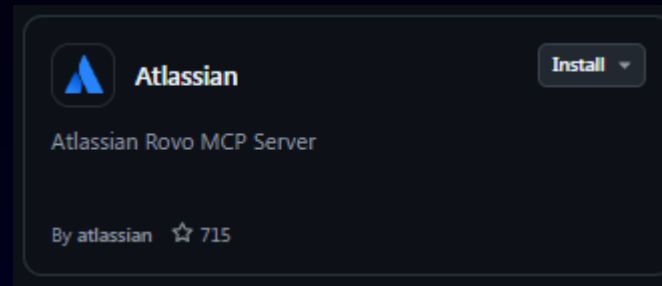
Add microsoft/azure-devops-mcp to mcp.servers

Create & update Work Items from Chat

Link PRs to Boards items automatically

Trigger Azure Pipelines via agent prompts

'What's in my current sprint?' — live backlog query

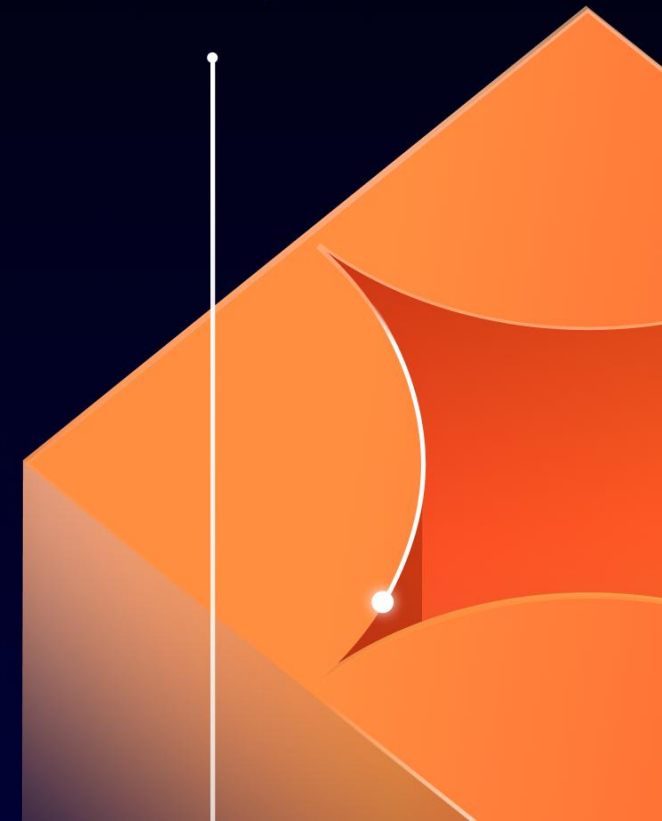


Jira

- Create Jira issues directly from Copilot Chat
- Update story points, status, and assignees
- Research a Jira epic before implementation
- 'Create a Jira story for the issue I just fixed

Confluence

- Auto-generate ADRs from research reports
- Publish PR summaries to your team wiki
- 'What does our wiki say about the auth flow?'





Thank you